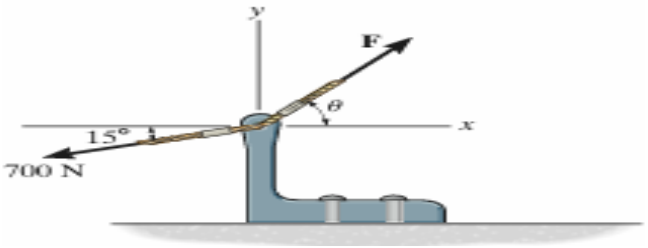
Assignment 1

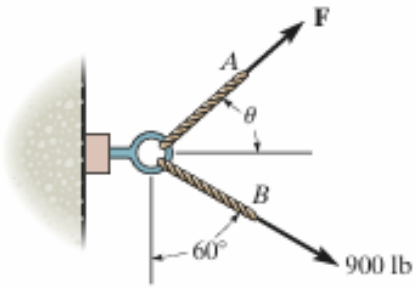
Question 1

If the magnitude of the resultant force is 500 N, directed along the positive y axis, determine the magnitude of force F and its direction θ.



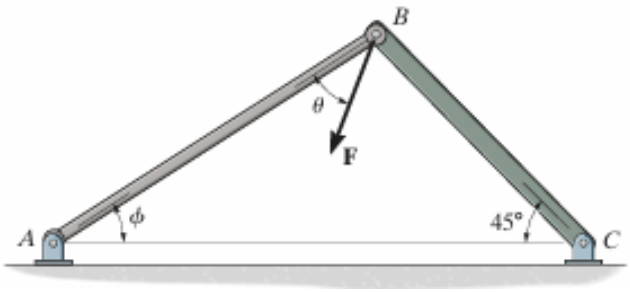
Question 2

If the resultant force acting on the support is to be 1200 lb, directed horizontally to the right, determine the force F in rope A and the corresponding angle θ.



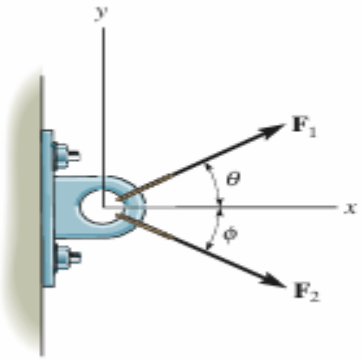
Question 3

Force F acts on the frame such that its component acting along member AB B is 650 lb, directed from towards ,and the component acting along member A BC from B C u is 500 lb, directed towards . Determine the magnitude of F and its direction . Set f = 60°



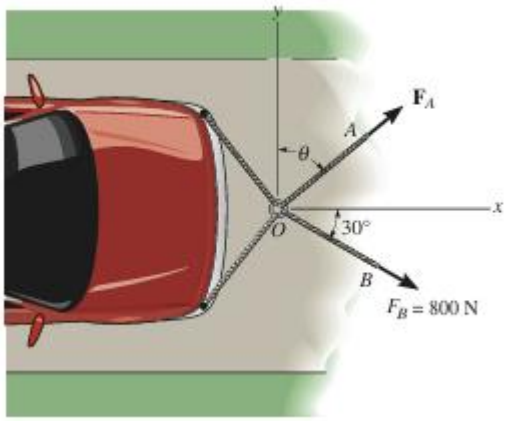
Question 4

If F1 = 30 lb and F2 = 40 lb, determine the angles u and f so that the resultant force is directed along the positive x axis and has a magnitude of FR = 60 lb



Question 5

Determine the magnitude and direction u of FA so that the resultant force is directed along the positive x axis and has a magnitude of 1250 N.



Question 6

Determine the magnitude of the resultant force FR = F1 + F2 and its direction, measured clockwise from the positive u axis

